

GOES-19 Full Disk Web Publisher

A deterministic, self-healing, production-grade pipeline for ingesting GOES-19 HRIT Full Disk imagery and publishing stable, continuously updating web endpoints.

This repository contains everything required to:

- Ingest GOES-19 HRIT (High Rate Information Transmission) • Decode ABI imagery • Select only **complete Full Disk frames** • Enforce data-settling guarantees (no partial files) • Publish stable web endpoints (`latest_*.png`) • Serve via nginx • Auto-heal via systemd watchdog • Provide deterministic updates • Provide a clean web UI

This is not a demo system — it is designed as **infrastructure**.

System Architecture

```
GOES-19 HRIT RF
  ↓
SDR + LNA + Filter + Dish
  ↓
SatDump HRIT Decoder
  ↓
Filesystem (timestamped frames)
  ↓
update_goes_fd_web.sh (publisher)
  ↓
/var/www/goes
  ↓
nginx
  ↓
Web UI + stable endpoints
```

Core Design Principles

1) Determinism

Only complete datasets are published. Partial frames are never visible.

2) Stability

Stable URLs:

```
/latest_false_color.png  
/latest_clean_ir.png  
/latest_longwave_ir.png  
/latest_wv_upper.png  
/meta.json
```

These never change names — only content.

3) Temporal Safety

Files must be **settled** before publication:

```
SETTLE_SECONDS=90
```

This prevents race conditions while SatDump is still writing.

4) Idempotence

Publisher can run repeatedly without corruption.

5) Self-Healing

A watchdog monitors freshness and automatically recovers.

Disk Layout

SatDump Output

```
/home/pi/sat/GOES-19/IMAGES/GOES-19/Full Disk/  
  2026-01-04_19-00-22/  
    G19_2_*.png   (CH2 Visible)  
    G19_7_*.png   (CH7 Clean IR / Shortwave IR)  
    G19_8_*.png   (CH8 Upper Water Vapor)  
    G19_13_*.png  (CH13 Longwave IR)
```

Each directory = one Full Disk frame timestamp.

Web Root

```
/var/www/goes/  
  index.html  
  meta.json  
  latest_false_color.png  
  latest_clean_ir.png  
  latest_longwave_ir.png  
  latest_wv_upper.png  
  current/
```

Channel Mapping

ABI Band	Meaning	Web Output
CH2	Visible	latest_false_color.png
CH7	Shortwave IR	latest_clean_ir.png
CH8	Upper Water Vapor	latest_wv_upper.png
CH13	Longwave IR	latest_longwave_ir.png

Publisher Logic

File:

```
/usr/local/bin/update_goes_fd_web.sh
```

Selection Algorithm

1. Sort timestamp directories (newest first)
2. For each directory:
3. Must contain CH2, CH7, CH8, CH13
4. Files must be older than `SETTLE_SECONDS`
5. First directory that passes → selected

This guarantees: • No partial frames • No torn writes • No mixed timestamps

Publication Actions

```
/meta.json           ← timestamp + UTC update time
/latest_false_color.png ← CH2
/latest_clean_ir.png  ← CH7
/latest_longwave_ir.png ← CH13
/latest_wv_upper.png  ← CH8
```

Ownership:

```
www-data:www-data
```

meta.json Format

```
{
  "timestamp_dir": "2026-01-04_19-30-22",
  "updated_utc": "2026-01-04T20:00:42Z"
}
```

This provides: • determinism • observability • automation hooks • validation

nginx Configuration

Primary Server

```
listen 8080;
root /var/www/goes;
```

Alias Compatibility

```
/goes/ → /var/www/goes/
```

This allows both:

```
/meta.json  
/goes/meta.json
```

Web UI

Features:

• Stable image endpoints • No caching • Auto refresh • Deterministic refresh • Labeled channels • Single-page interface

Captions:

• CH2 (Visible) — False color proxy • CH7 — Clean IR (3.9 μm) • CH13 — Longwave IR (10.3 μm) • CH8 — Upper-level water vapor

Timers

Publisher Timer

```
update-goes-fd-web.timer
```

Runs publisher automatically.

Watchdog Timer

```
goes-web-watchdog.timer
```

Functions:

• Detect stale data • Verify file ages • Verify meta freshness • Restart SatDump if needed • Re-publish if needed • Prevent restart storms • Enforce cooldowns

Watchdog Behavior Model

```
FRESH → do nothing  
STALE → publish  
STALE x N → restart SatDump  
COOLDOWN → wait  
RECOVER → reset counters
```

This is a **finite-state recovery machine**, not a cron script.

Why This Is Stable

• No filename mutation • No directory races • No partial visibility • No dependency on SatDump timing • No UI race conditions • No HTTP caching • No symbolic links • No symlink flips • No file renames

Only atomic copy + overwrite.

Operational Model

Deterministic until it fails, then adaptive.

Normal mode: • Deterministic publish cycle

Failure mode: • Detection • Isolation • Recovery • Stabilization • Resume

What This System Is

• Infrastructure • Data pipeline • Deterministic publishing system • Observability platform • Real-time geophysical data service

What This System Is Not

• Demo script • Hobby pipeline • "best effort" system • fire-and-forget cron job

Future-Proofing

Designed to support:

• Mesoscale sectors • Multi-sat relay • GOES-West integration • Himawari relay • EMWIN ingestion • DCS ingestion • Multi-product compositing • AI segmentation • Temporal differencing • Change detection • Motion vectors • Event detection

Installation Flow (Novice-Friendly)

1) Install SatDump 2) Configure HRIT for GOES-19 3) Set output directory:

```
/home/pi/sat/GOES-19/IMAGES/GOES-19/Full Disk
```

4) Install nginx

5) Deploy repo files:

```
/usr/local/bin/update_goes_fd_web.sh  
/etc/systemd/system/update-goes-fd-web.service  
/etc/systemd/system/update-goes-fd-web.timer  
/etc/systemd/system/goes-web-watchdog.service  
/etc/systemd/system/goes-web-watchdog.timer  
/var/www/goes/
```

6) Enable timers:

```
systemctl daemon-reload  
systemctl enable --now update-goes-fd-web.timer  
systemctl enable --now goes-web-watchdog.timer
```

7) Open browser:

```
http://<host>:8080/
```

Philosophy

This system is built on:

- Deterministic state machines • Observable invariants • Controlled mutation • Stable interfaces • Fault isolation • Minimal coupling • Predictable behavior

This is how infrastructure is built — not scripts.

Status

Operational

- Continuous publishing • Stable endpoints • Deterministic behavior • Self-healing • Full Disk only • No partial frames • Labeled channels • Production stability

Author's Note

This system intentionally prioritizes:

- correctness over speed • determinism over novelty • stability over features • invariants over convenience • architecture over hacks

Because once it's stable — everything else becomes easy.

License

Public repository.

Open architecture.

Free use for research, education, and infrastructure.

GOES-19 Full Disk Web Publisher

Built as infrastructure, not a project.